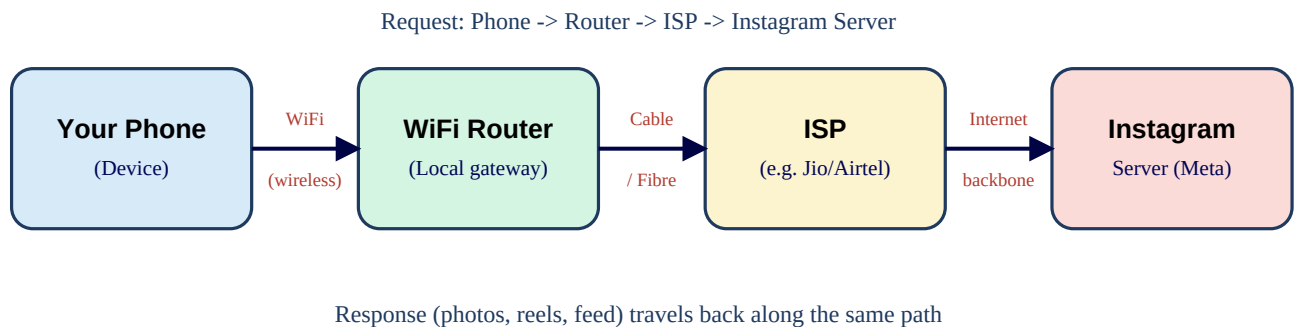


Introduction to Networks

Questions with Answers

1. Draw a simple diagram showing how your phone connects to Instagram (device, WiFi router, ISP, Instagram server). Label each part and the connections.

Answer:



- **Phone:** the end device; opens the Instagram app and sends the request.
- **WiFi router:** connects the phone wirelessly (WiFi) to the home/college network and forwards traffic to the ISP.
- **ISP:** the Internet Service Provider gives access to the wider Internet.
- **Instagram server:** a data-centre server owned by Meta that stores posts and sends back the feed.

2. List 5 devices you use or see daily that are part of a network, and write the type of network each connects to.

Answer:

Device	Network type	Explanation
Smartphone	LAN / WAN	Connects to home WiFi (LAN) and to the Internet/mobile data (WAN).
WiFi router	LAN and WAN	Creates the LAN at home and links it to the ISP's WAN.
Laptop	LAN	Connects through WiFi or an Ethernet cable inside a home/college.
Smartwatch	PAN	Pairs with the phone over Bluetooth within a few metres.
Smart TV	LAN	Connects over WiFi to stream Netflix/YouTube via the home network.

LAN = Local Area Network, WAN = Wide Area Network, PAN = Personal Area Network.

3. Explain the difference between the Internet and an intranet using an example from your college or a public WiFi you have used.

Answer: The **Internet** is a global public network of networks that anyone can access, while an **intranet** is a private network limited to one organisation and its authorised users.

Point	Internet	Intranet
Access	Open to everyone worldwide	Only college students/staff with login
Ownership	No single owner	Owned and managed by the college
Security	Less controlled, public	Firewall and login protected
Example	Google, YouTube, Instagram	College portal for attendance, results, e-library

Example: On the college WiFi, a student logs into the college portal to view results. That portal works only inside the college network (intranet). The same student then opens YouTube on the same WiFi, which goes out to the public Internet. On a public WiFi (e.g. at a café or railway station) there is no intranet, so everything is on the Internet and should be used carefully.

4. Pick any one app and describe in 4-5 lines how data travels from your device to the server and back when you search inside the app.

Answer (App: Zomato):

When I type "pizza" and tap **Search** in Zomato, my phone creates a request and sends it through the WiFi router (or mobile data) to my ISP. The ISP forwards it across the Internet, using DNS to find Zomato's server address, to Zomato's data-centre server. The server runs the search on its database of restaurants, finds matching results with prices and ratings, and packs them into a response. This response travels back along the same path (server, ISP, router, phone) in small data packets. The Zomato app on my phone then reassembles them and shows the restaurant list on screen, all within a second or two.

Devices and networks involved: phone, WiFi router (LAN), ISP, Internet (WAN), Zomato server in a data centre.